



Interply hybrid composites with carbon fiber reinforced polypropylene and self-reinforced polypropylene

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ABSTRACT

The focus of the present study is on hybrid composites with interplied carbon fiber reinforced polypropylene (CFRPP) between self-reinforced polypropylene (SRPP) layers. SRPP is produced by hot compaction of a woven fabric of highly oriented polypropylene and has an intrinsic behavior of shrinkage under high temperatures. The aim of this research is to enhance the tensile properties of the CFRPP/SRPP hybrid composites by using the SRPP shrinkage to introduce a compressive pre-strain in CFRPP. The results from tensile testing show that the failure strain of the hybrid composites is improved in comparison with CFRPP. The modulus and strength are noted to be lower than the ones expected from the rule of mixture. This may be attributed to the introduction of local misalignment (waviness) of carbon fibers caused by the SRPP shrinkage during consolidation.

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1. Introduction

Carbon fiber reinforced polymer composites have become widely used in many engineering fields where high mechanical properties and light weight are required. In recent years, thermoplastic polymers have attracted a lot attention as matrix material due to the benefit of omitting the curing process, to a less hazardous chemical composition and to its better recyclability. Among various polymer options, polypropylene (PP) is extraordinarily low density and cheap, thus preferably used in the automobile industry. Since carbon fibers are one of the strongest materials, carbon fiber reinforced PP (CFRPP) has a potential to achieve high specific modulus and strength.

In addition, self-reinforced PP (SRPP) is known to achieve much higher mechanical properties than conventional PP. Many researchers have worked on characterizing the processing and mechanical behavior of SRPP [1–8]. Propex Fabric commercialized SRPP sheet as Curv[®] [1], which is produced by hot compaction of a woven fabric made of highly oriented PP tapes. During hot compaction, the surface of the tapes is selectively melted and integrated together. The SRPP achieves excellent modulus, strength and impact resistance. Among other characteristics, SRPP has a tendency to shrink under high temperature [2,8], which is triggered by the entropy increase in a polymer packing state. If SRPP is heated under constrained conditions to prevent shrinkage, a recovery stress is generated. In this research, we aim to utilize the recovery stress by laminating SRPP with CFRPP in a hybrid composite.

The first hybrid composite reported in the literature was a composite combining together carbon and glass fiber. It exhibited increased failure strain in comparison with all carbon fiber composite [9]. The source of the hybrid effect was presumed to be a compressive residual stress in carbon fiber developed by the difference of thermal expansion coefficients of carbon and glass fiber [10]. The hybrid composite proposed in the present research is made of CFRPP and SRPP. The active shrinkage of SRPP should also apply a compressive pre-strain into the CFRPP, enhancing its tensile properties compared to CFRPP without SRPP.

This paper is organized as follows. Section 2 describes the preparation of CFRPP, which is made by PP film stacking impregnation of a woven carbon fabric. The influence of processing conditions on impregnation and tensile properties is investigated, and the conditions are optimized with respect to maximizing the tensile sustainable stress, while keeping the fiber volume portion constant. Section 3 is devoted to characterization of SRPP. The influence of processing conditions on tensile properties and recovery stress of SRPP is investigated. Section 4 presents the hybrid effect in the interply hybrid composite with CFRPP and SRPP. The hybrid composites are tested under tensile loading and their mechanical properties are compared with the ones expected from the rule of mixture.

2. Preparation of carbon fiber reinforced polypropylene

A prepreg made of continuous carbon fibers and PP was commercially unavailable as this research started. Thus, CFRPP was prepared by stacking a woven carbon fabric and PP film, and hot pressing them for impregnation.

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The fabric was twill 2/2 of HexForce® G0986 D1200 (Hexcel), woven with carbon fibers Tenax® HTA 5131 6 K (Toho Tenax). The tensile properties of this fiber were a Young's modulus of 238 GPa, a strength of 3950 MPa and a failure strain of 1.7% [11]. The PP film (Amdor), made of maleic anhydride modified PP, had a thickness of 100 μm , a melting temperature of 160 $^{\circ}\text{C}$ and a melt flow index (MFI) of 5.2 g/10 min.

The basic fabrication procedure for CFRPP was as follows. The carbon fabric cut in $300 \times 300 \text{ mm}^2$ square sheets was sandwiched between two PP films and the stack was then inserted between aluminum plates, with Teflon sheets between the stack and the plates; the aluminum plates were placed in a compression machine and hot pressed to impregnate PP into the fabric; after the aluminum plates were cooled down at the same pressure to consolidate the material; the CFRPP ply was demolded from the aluminum plates. For preparation of a CFRPP laminate, three CFRPP plies (aligned in warp direction) were stacked and consolidated together using the above procedure.

The impregnation was difficult to achieve because PP had to completely wet a large surface area of carbon fibers with a diameter of only 7 μm . Furthermore, the MFI of the PP film was too low to lead to perfect impregnation. Fig. 1a compares the cross-sections of well and poorly impregnated laminates. PP permeates the fibers in the well impregnated laminate (top), while at the center of the fiber bundles in the poorly impregnated laminate, a non-impregnated area colored in black can be seen (bottom). Fig. 1b compares the tensile fracture coupons of the well and poorly impregnated laminates. The well impregnated laminate was broken in a direction perpendicular to the tensile load – the same way as conventional woven carbon fiber/epoxy composites, except for the occurrence of fiber pull-out. The poorly impregnated laminate, on the other hand, was broken at an angle with a step-wise, irregular fracture surface including much more fiber pull-out.

The influence of processing conditions on the impregnation quality and on the tensile properties was investigated. The hot

pressing temperature, pressure and duration were selected as processing parameters (Table 1). In the fabrication procedure, the materials, inserted between aluminum plates, were placed in the compression machine at maximum 50 $^{\circ}\text{C}$ and then heated at 5 $^{\circ}\text{C}/\text{min}$ with minimum pressure of approximately 0.2 bar. When the materials reached the prescribed temperature, the prescribed pressure was applied for prescribed time. After completing the hot pressing, the plates were cooled down at 5 $^{\circ}\text{C}/\text{min}$ with the prescribed pressure maintained and finally demolded at below 50 $^{\circ}\text{C}$. The prescribed temperature was 180, 200 or 220 $^{\circ}\text{C}$; the prescribed pressure was 20, 30 or 40 bar; the prescribed duration time was 10, 30 or 50 min. The above conditions were applied not only for fabrication of CFRPP plies but also for consolidation of CFRPP plies into laminates.

To assess the impregnation quality of the obtained laminates, the fiber volume fraction (V_f) and the void volume fraction (V_v) were calculated from Eqs. (1) and (2), respectively,

$$V_f = \frac{\lambda_f N}{t_c \rho_f} \quad (1)$$

$$V_v = 1 - \frac{M_c}{A_c t_c \rho_m} + \frac{\lambda_f N (\rho_f - \rho_m)}{t_c \rho_f \rho_m} \quad (2)$$

where λ_f , N , M_c , A_c , t_c , ρ_f and ρ_m denote the areal weight of the carbon fabric, the number of stacked carbon fabrics, the laminate weight, the laminate area, the laminate thickness, the carbon fiber density, and the PP density, respectively. This study used the following values: $\lambda_f = 285 \text{ g/m}^2$, $N = 3$, $\rho_f = 1.76 \text{ g/cm}^3$, $\rho_m = 0.9 \text{ g/cm}^3$. Tensile testing was performed according to ASTM D3039. The load was applied in the warp direction. The modulus was calculated on the average slope of the stress–strain curves in the range of 0.1–0.3% strain. The number of specimens was five each. As summarized in Table 1, the influence of processing conditions on the void

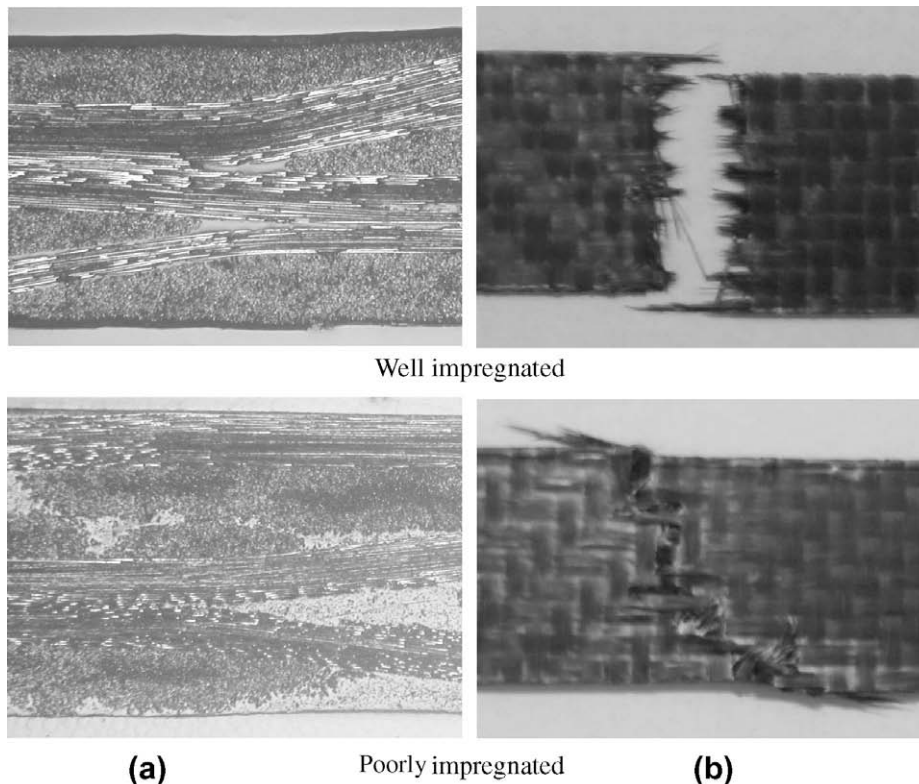


Fig. 1. Comparison of cross-sections (a) and tensile fracture coupons and (b) of CFRPP between well and poorly impregnated laminates.

Table 1

Tensile properties of CFRPP laminates made under different process conditions.

Code	Processing conditions			Geometrical properties			Modulus		Strength		Failure strain		Strength factor k^a
	Temperature (°C)	Pressure (bar)	Duration (min)	t_c (mm)	V_f (%)	V_v (%)	Ave. (GPa)	CV. (%)	Ave. (MPa)	CV. (%)	Ave. (%)	CV. (%)	
CFRPP 1	180	20	10	1.07	45.2	8.2	47.8	11	452	4	0.97	6	1.00
CFRPP 2	200	20	10	1.02	47.6	8.4	46.2	3	449	5	0.93	6	0.94
CFRPP 3	220	20	10	0.92	53.0	3.0	54.1	5	524	4	0.92	4	0.99
CFRPP 4	220	40	10	0.86	56.5	1.1	50.2	13	534	13	0.97	6	0.95
CFRPP 5	220	20	30	0.96	50.4	6.6	47.8	11	561	8	1.12	14	1.11
CFRPP 6	220	20	50	0.89	54.7	3.0	50.7	10	644	3	1.19	5	1.18
CFRPP 7	220	30	30	0.84	57.7	3.5	47.5	15	608	6	1.11	10	1.06
CFRPP 8	220	30	50	0.82	59.6	2.3	45.9	13	672	9	1.28	11	1.13

^a $k = (\text{strength}/V_f)/(\text{strength}/V_f)_{\text{CFRPP 1}}$.

content and tensile properties was investigated under eight processing conditions “CFRPP 1–8”.

Even though every laminate contained the same amount of carbon fabric and PP film, the laminate thickness t_c and the fiber volume fraction V_f varied with the processing conditions. More specifically, the thickness decreased and the fiber volume fraction increased as the temperature, pressure, duration time increased (individually or together). This was attributed to the improved impregnation quality and to the observation that PP was squeezed out from the aluminum plates. Fig. 2 illustrates the influence of the processing temperature on the void content and tensile strength under the same processing pressure (20 bar) and time (10 min). The void content dramatically decreased at 220 °C with simultaneous improvement of the tensile strength. Fig. 3 illustrates the influence of the processing pressure and time on the void content and tensile strength under the fixed processing temperature (220 °C). The void content decreased as the processing pressure became higher, but the strength only slightly decreased with the processing pressure. If a tensile sustainable stress per carbon fiber volume (tensile strength/ V_f) was normalized with that of “CFRPP 1” as a strength factor k depicted in Table 1, one can see that k apparently decreased as the processing pressure increased. A possible explanation for this is that high pressure caused compaction of a carbon fabric and restrained impregnation. On the other hand, longer processing times decreased the void content and improved strength. The comparison of the normalized sustainable stress (as

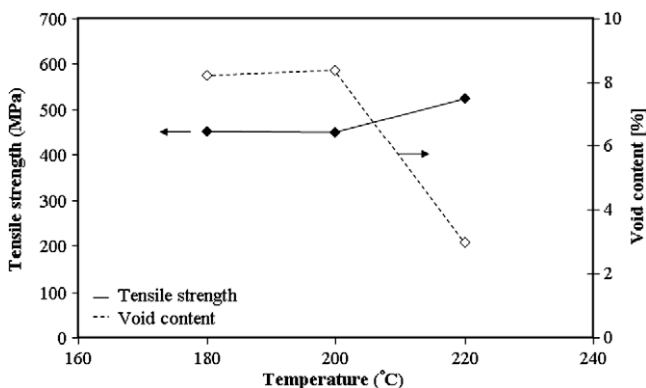


Fig. 2. Comparison of tensile strength and void content of CFRPP for various processing temperature under 20 bar processing pressure, 10 min processing time.

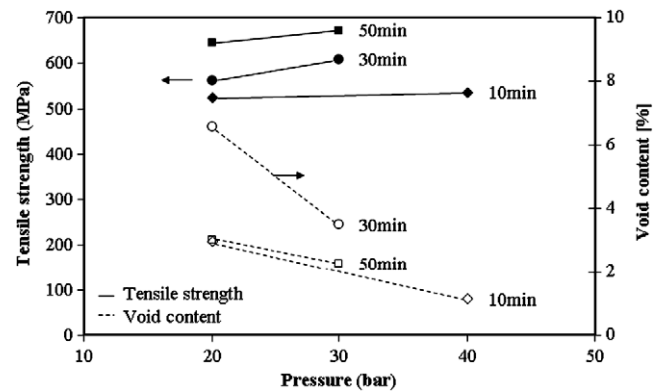


Fig. 3. Comparison of tensile strength and void content of CFRPP for various processing pressure and time under 220 °C processing temperature.

above) revealed that the combination of 220 °C, 20 bar, 50 min was the best processing condition. Therefore, we concluded that lower processing pressure is better as far as impregnation can proceed; higher processing temperature and longer processing time are also better as far as properties of PP are not degraded.

3. Characterization of self-reinforced polypropylene

This study used Curv® (Propex Fabric) with a thickness of 0.15 mm as a SRPP sheet. During the precompaction of the woven Curv® tapes, the surface of the tapes is molten and recrystallizes upon cooling. The melting temperature of this “matrix” part of Curv®, precompacted sheets is about 165 °C and that of remaining highly oriented PP part is about 180 °C.

First of all, the influence of the process conditions on the tensile properties of the SRPP laminate was investigated. The fabrication procedure for the SRPP laminates was as follows. SRPP sheets were cut into 300 × 300 mm² squares; six plies of them were all stacked in the warp direction and inserted between aluminum plates using Teflon sheets; then the stacks were placed in the hot zone in the compression machine pre-heated to a prescribed temperature and heated up with minimum pressure of approximately 0.2 bar; after reaching the prescribed temperature, hot pressing was applied under 20 bar for 10 min; the stacks were then quickly transferred to the cold zone in the compression machine under room

Table 2
Tensile properties of SRPP laminates.

Processing temperature	Modulus (GPa)	Strength (MPa)	Failure strain (%)
165 °C	2.8	149	21.7
170 °C	2.5	147	22.5
175 °C	2.5	142	24.1

temperature for cold pressing. The heating and cooling rates were both about 25 °C/min. The prescribed temperature was 160, 165, 170, 175 or 180 °C.

The SRPP laminate produced at the processing temperature of 160 °C was not completely consolidated and that produced at 180 °C was melted and squeezed out from the aluminum plates. As summarized in Table 2, the other SRPP laminates were tested in tension according to ASTM D3039. The tensile load was applied in the warp direction. The initial modulus was determined on the average slope of the stress–strain curves in the range of 0.1–0.3% strain. The number of specimens are five each. Fig. 4 illustrates the non-linear stress–strain curves of the SRPP laminates produced at the processing temperature of 165, 170 or 175 °C. The modulus and strength improved as the processing temperature became lower. On contrary, the failure strain improved as the processing temperature became higher as shown in Table 2. Therefore, we concluded that the lowest processing temperature (165 °C) is the best for maintaining the potential of the highly oriented PP, as far as a perfect consolidation can be achieved.

Secondly, the recovery stress of SRPP that is expected to be the driving force for the hybrid effect was estimated. The recovery stress was estimated from the curvature of an unbalanced laminate of [SRPP₆/CFRPP₃] produced by consolidating the SRPP laminate prepared by the above mentioned procedure and the CFRPP laminate. Fig. 5 explains the method to determine the recovery stress of SRPP. The cross-section of an unbalanced laminate with SRPP and CFRPP (Fig. 5a) shows the equilibrium of force induced by bending of the unbalanced laminate and the recovery stress of SRPP. In this calculation, the residual stress induced by the difference of thermal expansion coefficient between CFRPP and SRPP was deliberately included into the recovery stress of SRPP. The equilibrium force and moment about the neutral axis can be expressed as the following equations:

$$\int_{-a-b+c}^{-b+c} (\sigma_r + E_a \frac{y}{R}) dy + \int_{-b+c}^c E_b \frac{y}{R} dy = 0 \quad (3)$$

$$\int_{-a-b+c}^{-b+c} (\sigma_r + E_a \frac{y}{R}) y dy + \int_{-b+c}^c E_b \frac{y^2}{R} dy = 0 \quad (4)$$

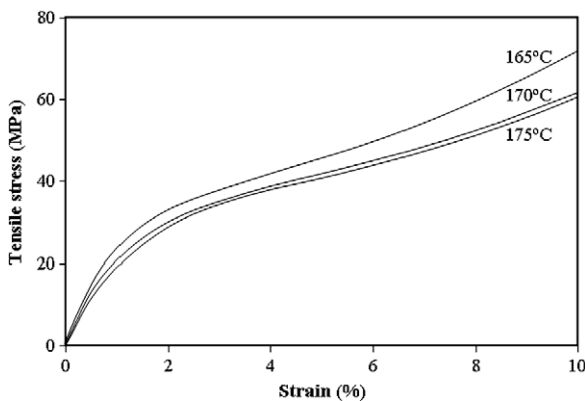


Fig. 4. Stress–strain curve of SRPP laminate consolidated under 20 bar processing pressure, 165 °C processing temperature, 10 min processing time.

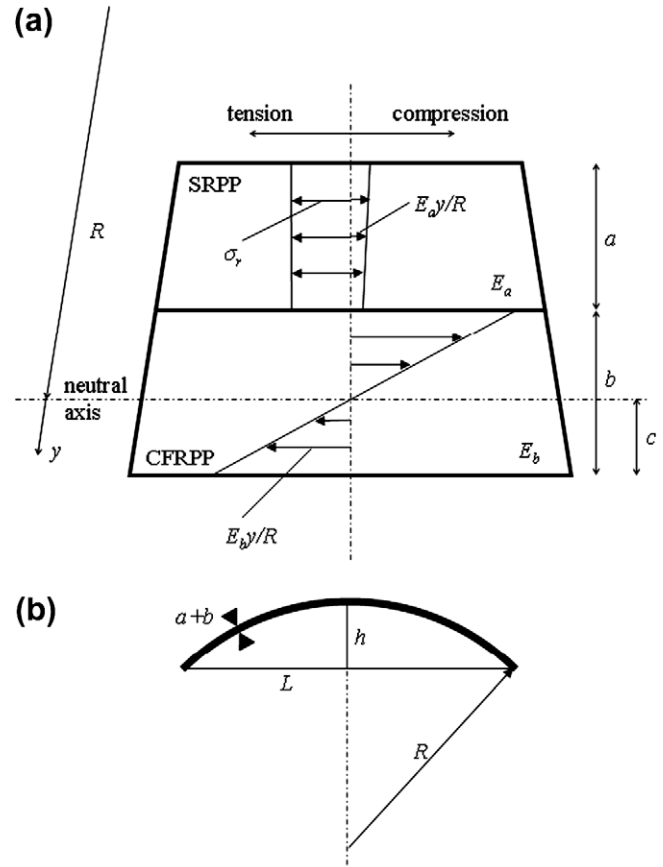


Fig. 5. Method for determining recovery stress of SRPP: (a) equilibrium of force and moment about neutral axis and (b) deformation of unbalanced laminate.

where σ_r , a , b , c , E_a , E_b and R denote recovery stress of SRPP, thickness of SRPP, thickness of CFRPP, distance between the CFRPP surface and the neutral axis, SRPP modulus, CFRPP modulus and curvature radius to the neutral axis respectively. Combining Eqs. (3) and (4) yields σ_r as written in Eq. (5),

$$\sigma_r = \frac{(E_a a^2 - E_b b^2)^2 + 4E_a E_b a b (a + b)^2}{6E_a b (a + b) R} \quad (5)$$

E_a , E_b were experimentally measured as 3.8 GPa, 35 GPa, respectively, by three point bending tests of SRPP laminate and CFRPP laminate before fabricating the unbalanced laminate. The value of R can be calculated from Eq. (6):

$$R = \frac{1}{2} \left(\frac{L^2}{4h} + h \right) \quad (6)$$

where h and L are height and projected length of the unbalanced laminate placed on a flat plate, respectively (Fig. 5b).

The influence of the processing conditions on the recovery stress of SRPP was investigated. The fabrication procedure for the unbalanced laminate was the same as mentioned for fabrication of SRPP laminate with exception of the prescribed temperature, which in this case was 160, 170 or 180 °C. A strip of specimen (250 × 25 mm) was cut out from the center part of each obtained unbalanced laminate and the curvature was measured in the longitudinal direction. The estimated recovery stress under these processing temperatures is summarized in Fig. 6. It is clear that there is an optimal condition for maximizing the recovery stress

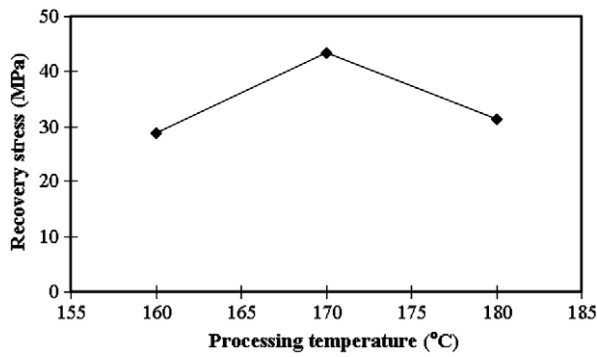


Fig. 6. Recovery stress of SRPP under various processing temperature.

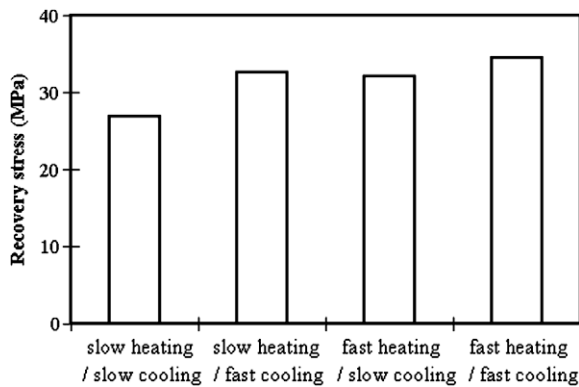


Fig. 7. Recovery stress of SRPP for various heating and cooling rates.

in the range of 160–180 °C. The peak recovery stress was assumed to be approximately 40 MPa.

The influence of the heating and cooling rates was also investigated. In the same processing procedure, the prescribed temperature was fixed at 165 °C. The heating and cooling rates were both applied at 5 °C/min and 25 °C/min as respectively “fast” and “slow” rate. The hot pressing time was set for 10 min except that the shorter processing time of 1 min was applied to the combination of fast heating and cooling to check the influence of exposure time at the high temperature and pressure. A strip of specimen (250 × 25 mm) was cut out from the center part of each obtained unbalanced laminate and the curvature was measured in the longitudinal direction. As summarized in Fig. 7, the combination of the fast heating and cooling and the short processing time achieved the highest recovery stress. This result revealed that a shorter time for exposing the laminate to a high temperature brings a higher recovery stress. We hypothesized that the recovery stress induced by the shrinkage of SRPP was partially relaxed under high temperature during the laminate fabrication.

Consequently, we concluded that the minimum processing temperature is suitable for maintaining tensile properties of SRPP as far as SRPP can be consolidated with CFRPP; shorter processing time at high temperature is effective for maximizing the recovery stress of SRPP.

4. Demonstration for interply hybrid composites

Interply hybrid composites were produced by alternatively laminating CFRPP and SRPP. Stacking sequences of prepared specimens were [SRPP₂/CFRPP/SRPP₂] as Hybrid 1, [CFRPP/SRPP]_s as Hybrid 2 and [SRPP/CFRPP/SRPP/CFRPP/SRPP] as Hybrid 3, where both SRPP and CFRPP were aligned in the warp direction, which was also the

loading direction in tensile tests. The fabrication procedure, basically referred to in Sections 2 and 3, was as follows. First the carbon fabric cut in 300 × 300 mm² square sheet was sandwiched between two PP films and then inserted between aluminum plates, with Teflon sheets; the stack was placed in a compression machine at maximum 50 °C and heated at 5 °C/min; after reaching 220 °C hot pressing was applied under 20 bar for 50 min; the stack was cooled down at 5 °C/min with the pressure kept and the CFRPP ply was finally demolded at below 50 °C. The obtained CFRPP plies and SRPP sheets cut in 300 × 300 mm² squares were stacked in the prescribed stacking sequence and again inserted between the aluminum plates with Teflon sheets; the stacks were set in the compression machine pre-heated to 165 °C; after reaching 165 °C hot pressing was applied under 20 bar for 10 min; the stacks were then transferred for cold pressing to the cooler zone of the compression machine with the pressure plates at room temperature; consequently the interply hybrid composite was demolded.

Fig. 8 shows the cross-section of one of the hybrid composites “Hybrid 3”. The void content of the hybrid composites increased compared with the CFRPP laminate. This is explained by the lower temperature used for consolidation of CFRPP and SRPP than that for consolidation of CFRPP plies as depicted in Section 2. The hybrid composites were tested in tension according to ASTM D3039 as explained above.

Table 3 compares the tensile properties of the hybrid composites with those of the CFRPP and the SRPP laminates. The tensile modulus and strength of the hybrid composites were predicted by the rule of mixture. For this calculation, the tensile stress of SRPP (25 MPa) at the failure strain of CFRPP (1.19%) was extracted from the non-linear stress–strain curve of SRPP in Fig. 4 and applied to the SRPP component in the rule of mixture for strength. The results show that the failure strain of the hybrid composites is improved by 0.08–0.22% strain in comparison with the CFRPP. The modulus and strength, however, were found to be somewhat lower than the ones predicted from the rule of mixtures. We conclude that a hybrid effect could only be found in terms of failure strain, which increased by 7–18% in comparison with the failure strain of the non-hybridized CFRPP.

As summarized in Table 4, the compressive pre-strain induced by the shrinkage of SRPP in each specimen was calculated from the rule of mixture, where the recovery stress of SRPP was assumed to be 40 MPa. The results showed that the pre-strain was 0.11–0.21%, which reasonably agreed with the improvement of the failure strain. However, this could not explain the decrease of modulus and strength. We hypothesized that the recovery stress generated by the shrinkage of SRPP layers during processing locally disturbed alignment of carbon fiber in CFRPP layers and led to a reduced stiffness (–18% to –25%). Even at an increased failure strain, this led to a small reduction in tensile strength (–5% to 9%). The increase rate of failure strain always exceeded the reduction rate of

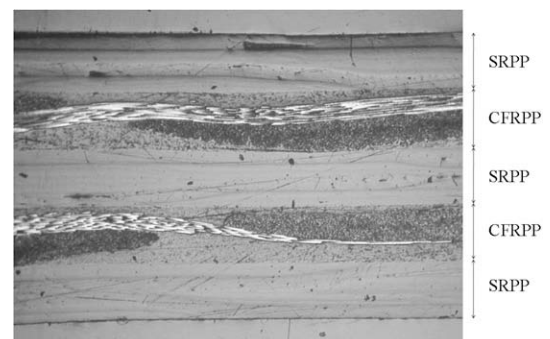


Fig. 8. Cross-section of interply hybrid composite “Hybrid 3”.

Table 3

Tensile properties of interply hybrid composites for experiments and predictions.

Code	Stacking sequence	Geometrical properties			Modulus		Strength		Failure strain
		t_c (mm)	V_f (%)	V_v (%)	Experiment (GPa)	Expected (GPa)	Experiment (MPa)	Expected (MPa)	Experiment (%)
CFRPP 6	[CFRPP] ₃	0.89	55	3	50.7	–	644	–	1.19
SRPP 1	[SRPP] ₆	0.91	0	–	2.8	–	149	–	21.7
Hybrid 1	[SRPP ₂ /CFRPP/SRPP ₂]	0.96	17	3	15.4 (18%↓)	19	219 (5%↓)	231	1.27 (7%↑)
Hybrid 2	[CFRPP/SRPP] ₅	0.99	33	5	26.3 (24%↓)	35	398 (9%↓)	438	1.41 (18%↑)
Hybrid 3	[SRPP/CFRPP/SRPP/CFRPP/SRPP]	1.14	28	4	22.8 (25%↓)	30	347 (8%↓)	378	1.31 (10%↑)

Table 4

Comparison of theoretical compressive pre-strain and improved failure strain among various interply hybrid composites.

Code	Stacking sequence	Volume fraction of SRPP (%)	Theoretical pre-strain (%)	Improved failure strain (%)
Hybrid 1	[SRPP ₂ /CFRPP/SRPP ₂]	66	0.21	0.08
Hybrid 2	[CFRPP/SRPP] ₅	33	0.11	0.22
Hybrid 3	[SRPP/CFRPP/SRPP/CFRPP/SRPP]	43	0.13	0.12

strength. Since the stress–strain curves are almost linear, the hybrid composites could improve the material potential in terms of strain energy.

5. Conclusions

In this research, interply hybrid composites with carbon fiber reinforced polypropylene (CFRPP) and self-reinforced polypropylene (SRPP) were investigated. SRPP has an intrinsic behavior of shrinkage under high temperature, and hence could induce a compressive pre-strain in the carbon fiber containing CFRPP layers during consolidation of the CFRPP/SRPP hybrid composite. Therefore, we expected a positive hybrid effect on the tensile failure, which is anyway dominated by the failure strain of the carbon fibers.

First, CFRPP was prepared by PP film stacking impregnation of carbon fabric. The influence of the processing conditions on the impregnation quality and the tensile properties was investigated, optimizing the conditions with respect to maximizing tensile sustainable stress per carbon fiber volume. Second, SRPP produced by hot compaction of highly oriented PP woven fabric was characterized. The influence of processing conditions on tensile properties and recovery stress was investigated. Finally, the interply hybrid composites with CFRPP and SRPP were produced under the optimized condition. Tensile testing of the composites revealed that the failure strain improvement was in reasonable agreement with the compressive pre-strain induced by SRPP shrinkage. The modulus and strength, however, were found to be lower than the ones predicted from the rule of mixture. Local carbon fiber misalignment caused by the SRPP shrinkage during the consolidation may reduce the stiffness and hence to hinder the expected hybrid effect. However, in terms of failure strain, a positive hybrid effect of +7% to +18% could be obtained. Further investigations are being carried out to evaluate the effect of the compressive pre-strain on other damage tolerance characterization like the strain to matrix crack

initiation in tensile and fatigue loading, and the impact resistance. Results of these studies will be reported in a next paper.

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